

Windows Subsystem for Linux (WSL)

Intro. to Computing

<https://mandarm.github.io/computing.html>

Installation

- Open Windows PowerShell (WPS).
- Run `wsl --install`.
 - If it says Downloading: Ubuntu, type Control-C to abort.
- Run `wsl --update`.
(should say most recent version is already installed)
- If you do not already have a copy of <https://releases.ubuntu.com/26.04/ubuntu-26.04-wsl-amd64.wsl>, download it.
- Double-click the file to launch the installer, OR run
`wsl --install --name ubuntu26 -from-file .\ubuntu-26.04-wsl-amd64.wsl`
- Close the window or type Control-D to exit.
- Run `wsl` in WPS to start WSL again.

Post installation tasks

- Run `wsl -l` to see what versions of Ubuntu are installed.
- If necessary, run `wsl --set-default`.
- Start `wsl` and run
`sudo apt update && sudo apt upgrade`.
- To choose a different version, run `wsl -d <version name>`.

■ How do Windows locations appear in WSL and vice versa?

- `C:\` $\equiv$ `/mnt/c`
- `/` $\equiv$ `\\wsl.localhost\Ubuntu-26.04` or `\\wsl$` (in Windows File explorer address bar)

■ **General guideline:** for best performance, avoid working across environments (Windows and WSL)

- When programming in WSL, store files under `/home/student` (or `/home/cssc ...`)
not `/mnt/c/Users/CSSC/Project` or `C:\Users\CSSC\Project`
- Exception (for now): edit your Python programs using Windows; run them using WSL.

■ **Warning re: file/directory names:**

Windows $\rightarrow$ case-insensitive, Linux $\rightarrow$ case-sensitive

Mixing Windows and WSL commands

- Using Windows executables from WSL: include .exe extension

Examples:

```
explorer.exe .
```

```
notepad.exe .bashrc
```

```
notepad.exe .bash_aliases
```

```
notepad.exe "C:\temp\foo.txt"
```

```
notepad.exe C:\\temp\\foo.txt
```

BUT `notepad.exe ~/.bash_aliases` **does not work!**

- Using Linux programs from Windows: add `wsl` before command name

Examples:

```
wsl ls -l
```

References

Official docs: <https://learn.microsoft.com/en-us/windows/wsl>